Smart IT Support Ticket Classifier – Project Proposal & Execution Plan

1. Project Objective

Build an NLP-based application that automatically classifies IT support tickets into their corresponding topic groups. The project demonstrates NLP, machine learning, evaluation, and deployment skills.

2. Problem Statement

Organizations receive thousands of support tickets daily. Manually categorizing them is time-consuming. The system predicts the ticket category from the textual description.

3. Input and Output

Input: Ticket Text.
Example: 'I cannot access shared storage.'
Output: Predicted Topic Group = Storage, Assigned Team = Storage Team.

4. Dataset

Dataset: IT Service Ticket Classification Dataset.
Columns used:
• Document (ticket text)
• Topic_group (target variable).

5. Business Perspective

Ticket Text → Topic Prediction → Team Assignment → Faster Resolution.

6. NLP Concepts

Text Cleaning, Tokenization, Stopword Removal, Lemmatization, TF-IDF, Text Classification.

7. Algorithms

Logistic Regression, Multinomial Naive Bayes, Random Forest.

8. Evaluation

Accuracy, Precision, Recall, F1 Score, Classification Report, Confusion Matrix.

9. Team Routing

Storage → Storage Team
Hardware → Hardware Team
Access → Access Team

Administrative Rights → Admin Team.

10. Streamlit Application

User enters a ticket and receives Topic Prediction, Confidence Score and Assigned Team.

11. Execution Plan

Phase 1: EDA
Phase 2: Preprocessing
Phase 3: TF-IDF
Phase 4: Model Training
Phase 5: Evaluation
Phase 6: Save Model
Phase 7: Streamlit Deployment
Phase 8: GitHub Documentation.

12. Expected Skills

NLP, TF-IDF, Scikit-Learn, Classification, Streamlit, End-to-End ML Workflow.

13. Future Scope

Embeddings → Vector Databases → LangChain → RAG → Resume-JD Matcher.